

ENGINEERING GEOLOGY

ROCK MECHANICS

ARABIAN SHIELD

Najd Fault System: Engineering Geology of the Arabian Shield



A technical look at the Najd Fault System in western Saudi Arabia—the largest Proterozoic shear zone system on Earth—and its first-order control on rock-mass engineering behavior, classification indices, and the performance of future major infrastructure projects in the region.

Document Structure & Scope

This document presents a structured technical narrative linking Najd tectonics to rock-mass engineering performance. It is organized to guide the reader from regional geological setting through structural mechanics, rock-mass classification, and kinematic analysis, culminating in a practical design workflow applicable to slopes, tunnels, deep foundations, and underground facilities in Najd-affected crystalline terrains.

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Introduction & Regional Setting	Tectonic Characteristics	Seven Mechanisms of Engineering Influence
Pan-African evolution, tectonic context, and geographic extent of the Najd system across the Arabian–Nubian Shield.	Key structural attributes — orientation, kinematics, displacement, and mechanical regime variability.	Expanded treatment of how Najd structures control block size, discontinuity sets, GSI/RMR, groundwater, and failure modes.
04	05	
Rock-Mass Anisotropy	Design Workflow & Zonation	
Structural, mechanical, and hydraulic anisotropy induced by the Najd shear fabric and its directional design implications.	Investigation strategy, Najd-intensity zoning, traffic-light slope classification, and support-level assignment.	

1. Introduction & Regional Geological Setting

The Najd Fault System is the largest Proterozoic shear zone system on Earth, forming a NW–SE striking belt more than 1,100–1,200 km long and up to 300 km wide across the northern Arabian Shield. It consists of a braided network of left-lateral strike-slip faults, ductile shear zones, and brittle fault strands that were active mainly between approximately 630 and 530 Ma, during the late stages of Pan-African evolution of the Arabian–Nubian Shield. This strike-slip system overprinted earlier arc-accretion structures and exerted a first-order control on crustal segmentation, exhumation of basement domes, and emplacement of igneous and hydrothermal bodies across Saudi Arabia and Egypt.

In western Saudi Arabia, Najd-related fabrics cut granitic and metavolcanic rocks that today form the foundation for many large-scale engineering projects, including transportation corridors, mega-developments, and strategic underground works. From an engineering geology perspective, the Najd system is therefore not only a tectonic feature but also a regional predictor of rock-mass quality, groundwater behavior, and failure mechanisms along planned project alignments. Understanding the tectonic imprint of the Najd system is a prerequisite for defensible geotechnical characterization of any site within the western Arabian Shield.

2. Tectonic Characteristics Relevant to Engineering

Regional-Scale Attributes

Najd structures are dominantly NW–SE trending, subvertical to steeply dipping, and show cumulative strike-slip displacements exceeding 200 km at the scale of the entire belt. Field and microstructural studies show evidence for both transpressional and transtensional regimes along different segments, leading to local uplift and exhumation of high-grade metamorphic rocks in some domains and extensional basins in others.

At the outcrop scale, Najd structures appear as brittle-ductile shear zones, fault gouge zones, cataclasites, and dense joint and fracture corridors commonly accompanied by minor grabens, folds, and dike swarms. This mechanical complexity explains the strong spatial variability of rock-mass conditions.

Key Engineering Attributes

- **Regional persistence:** Preferred NW–SE orientation of shear zones extends for hundreds to thousands of kilometers
- **Steep dip geometry:** Subvertical to steep dips favor structurally controlled block and wedge failures in high cuts
- **Kinematic reversal history:** Early dextral, later sinistral reactivation further damaged pre-existing discontinuities
- **Spatial variability:** Intact, high-quality basement may occur only hundreds of meters from intensely sheared, poor-quality zones
- **Structural overprinting:** Najd fabric is superimposed on earlier arc-accretion structures, compounding discontinuity complexity

Rock-Mass Fragmentation into Blocks of Varying Sizes

Najd deformation cuts originally massive crystalline rocks into smaller blocks, fundamentally lowering rock-mass strength and stability. At the belt scale, strike-slip shearing created kilometer-scale crustal blocks bounded by major faults, while at project scale it generated dense networks of secondary joints and fractures that control block size in slopes, caverns, and foundations. In many Najd corridors, cataclastic zones several meters to tens of meters wide appear as bands of very low RQD, often with block sizes ranging from decimeters to a few centimeters. This leads to a sharp local drop in GSI relative to adjacent, less-sheared rock.

Block size reduction is commonly anisotropic: blocks are elongated parallel to the shear direction (NW–SE), which creates directional differences in deformability and stability when slopes or tunnels intersect these fabrics at different orientations. For design purposes, distinguishing "background" jointing from "Najd-enhanced" fragmentation is critical for zoning the rock mass into different support classes and for locating structures away from the worst corridors where feasible.

Belt Scale

Kilometer-scale crustal blocks bounded by major strike-slip fault strands

Project Scale

Dense secondary joint and fracture networks; RQD reduction from >80% to <25% over short distances

Cataclastic Zones

Meter-wide bands of centimeter-to-decimeter block sizes requiring highest support classes

Systematic Discontinuity Sets and Shear Zones

The Najd system produces a predictable structural fabric dominated by NW–SE shear zones and associated conjugate sets trending NE–SW or N–S. At the project scale, this translates into several persistent joint and fault sets with specific orientations and spacing, which can be mapped and incorporated into kinematic analyses for slopes and underground excavations. The concept of structural anisotropy is central here: rock-mass stiffness, strength, and permeability all become direction-dependent because of the alignment of discontinuities with the Najd trend.

Riedel shears, tension gashes, and en-echelon veining within Najd zones introduce additional minor sets that may not appear in regional maps but are important in local wedge formation and seepage pathways. The persistence of some Najd faults exceeds several hundred kilometers, which means that once a problematic joint set is identified at one project site, the same orientation is likely to be critical for other projects along strike — a powerful implication for regional planning.

Field Evidence: Joint-Set Statistics

Combined measurement data from the Makkah region reveal the following dominant orientations consistent with Najd structural heritage:

- **120–130° strike bin (82° mean dip):** Strong Najd-parallel candidate — steeply dipping, high persistence
- **130–140° strike bin (89° mean dip):** Najd-parallel candidate — near-vertical, 100% steep percentage
- **140–150° strike bin (40° mean dip):** Najd-parallel candidate — moderately dipping, notable in wedge analysis
- **40–60° strike bins:** Conjugate or oblique sets associated with Najd transpression

Engineering Implication of Persistence

Once a Najd-parallel joint set is confirmed at a project site, the regional structural record supports its extrapolation along strike to other planned segments. This allows:

- Early screening of corridor segments for structural risk before detailed investigation
- Justification for conservative GSI/RMR assumptions along mapped Najd corridors
- Optimized borehole placement targeting confirmed high-intensity zones
- Stereonet analysis integrating both site-specific and regional structural data

Reduction of Rock-Mass Quality: GSI and RMR

Najd deformation reduces rock-mass quality through three reinforcing mechanisms: decreased block size, poorer joint surface conditions, and enhanced groundwater influence. In the core of major Najd shear zones, intense shearing produces fault gouge, breccia, and strongly weathered rocks with slickensided, clay-coated surfaces, which significantly lower joint friction angles and cohesion. The transition from intact basement to highly sheared material is often abrupt over a few meters, so GSI and RMR can change by one or two classification classes over very short distances — a critical finding for tunnel support design and slope stabilization.

Fresh Granite / Metavolcanics

GSI: 60–80

RMR: 60–75

Massive to blocky structure, rough planar joints, dry to moist conditions. Good to very good rock class. Light support sufficient in tunnels.

Moderately Sheared Najd Zone

GSI: 35–55

RMR: 35–55

Blocky/disturbed structure, slightly weathered joint walls, clay coatings developing. Fair rock class. Systematic support required.

Core of Najd Shear Corridor

GSI: 10–30

RMR: 15–35

Crushed/sheared with gouge and breccia, slickensided surfaces, clay infill, wet to flowing. Poor to very poor rock. Heavy support essential.

⚠ These GSI/RMR ranges are conceptual and illustrate the typical degradation from fresh crystalline rock to strongly sheared Najd core. They are not site-specific design values and must be verified through field investigation and core logging.

Groundwater Flow and Weathering Control

Najd faults function as major groundwater conduits, and this hydraulic role has profound implications for future deep infrastructure. Regionally, Najd shear and fracture corridors guide fluid flow, have been associated with hydrothermal mineralization, and govern modern groundwater circulation and weathering patterns. Fault cores composed of gouge and breccia can exhibit high permeability initially; with time, mineral precipitation and clay development may reduce permeability locally but concentrate flow along damage zones and fracture corridors parallel to the fault strike.

Enhanced weathering along these preferential flow paths leads to "weathering chimneys" — vertical zones of deeply altered rock that may intersect future foundations or tunnel crowns, acting as weak, deformable columns within an otherwise competent rock mass. Najd-controlled groundwater can create elevated pore pressures along specific joint sets, reducing effective normal stress and promoting sliding, especially where seepage cannot be efficiently drained from steep rock slopes or buried fault zones beneath heavy foundations.



Hydraulic Conductivity

Permeability is significantly higher parallel to NW–SE Najd shear zones and lower perpendicular to them, creating strongly anisotropic groundwater flow fields that must be characterized for drainage design.



Weathering Chimneys

Deep alteration fronts propagate preferentially along Najd structures, producing vertically irregular weathering profiles that invalidate simple depth-based assumptions for foundation design.



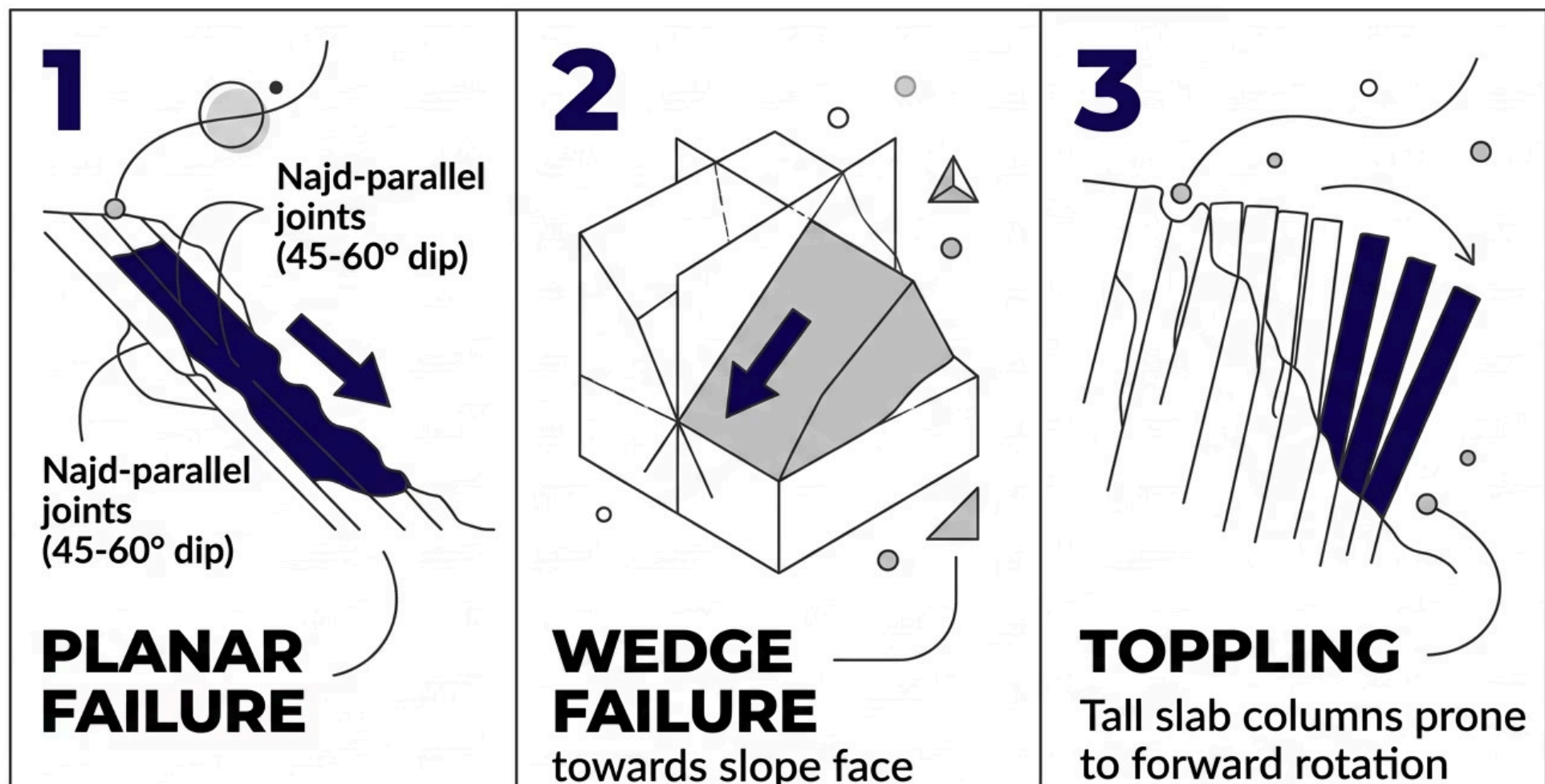
Pore-Pressure Effects

Elevated pore pressures along Najd-parallel discontinuities reduce effective normal stress, critically lowering the factor of safety for planar and wedge failures even under moderate static loading.

Planar and Wedge Failure Likelihood

The kinematic conditions for planar and wedge failures are directly tied to Najd geometry. Steep NW–SE trending slopes parallel to Najd lineaments often intersect one dominant joint set roughly parallel to the slope face and another oblique set, creating large wedges whose line of intersection daylights toward the free face. In cuts oriented perpendicular to the Najd trend, major shear zones may daylight as continuous, low-strength planes dipping out of the slope, controlling large planar slides and necessitating heavy reinforcement or re-orientation of the slope batter.

Rockfall hazard frequently concentrates in locations where minor Riedel or cross-faults intersect cut faces, locally undermining block support. For a highway cut oriented NE–SW across a NW–SE Najd corridor, stereonet analysis typically reveals one major joint set dipping 45–60° toward the cut, with a secondary set forming wedges that daylight at the slope face — a combination consistent with recurrent rockfall incidents observed along existing roads in the western Shield. Systematic scanline mapping and stereographic analysis of these sets should therefore be standard practice in all Najd domains.



Excavation Stability, Tunneling, and Foundations

Najd-related anisotropy leads to highly directional factors of safety in open cuts. Slopes parallel to the shear zone strike may show acceptable stability, while those cut obliquely can experience rapid degradation, overbreak, and progressive failure, requiring differential support strategies along the same project alignment. For underground works, intersections with major Najd shear zones are likely to coincide with poor-rock classes exhibiting high deformability, overbreak, block falls, and water inflow, while intervening segments in less-sheared rock may behave as good rock with minimal support needs.

Open-Cut Slopes

Najd-related anisotropy creates directional FS variability along a single alignment. Slopes cut obliquely to Najd corridors may experience rapid overbreak and progressive failure. Differential support strategies — heavier in Najd-intense zones, lighter in quiet blocks — are required even within short project segments.

Tunnel Excavations

Crossings of Najd shear corridors should be anticipated as "poor rock" reaches with high deformability and water inflow. The observational method with flexible support classes is recommended. Pre-support measures — forepoling, spiling, consolidation grouting — should be planned at all mapped Najd corridor crossings before breakthrough.

Shallow Foundations

Differential settlement risk is elevated where footings span the transition between sheared and relatively intact blocks. Bearing capacity and modulus contrasts of an order of magnitude can occur over plan distances of a few meters in Najd corridors, requiring continuous foundation investigation and load distribution analysis.

Deep Foundations / Piles

Variable shaft resistance is expected where piles traverse multiple Najd-related units. Piles should ideally avoid seating in thick gouge zones where long-term creep and strain localization may occur. Down-drag forces should be assessed where piles pass through deeply weathered Najd-parallel zones.

Najd System as a Regional Predictor of Rock Performance

The most powerful application of the Najd structural record is its use as a first-order predictor of rock-mass engineering performance before any project-specific investigation is conducted. Remote sensing tools — satellite lineament analysis, LiDAR, and InSAR — combined with existing 1:250,000 and 1:500,000 geological maps can delineate Najd structural intensity at the regional scale. High lineament density and strong NW–SE trends flag zones where lower GSI/RMR and more complex kinematics are anticipated, directing investigation resources to the most critical areas and away from structurally quiet basement blocks.

This predictive approach links regional Najd zoning directly to standard rock-mechanics tools: structural intensity guides preliminary GSI/RMR assumptions, which feed Hoek–Brown parameters, which feed limit-equilibrium and numerical analyses for slopes and tunnels, producing first-order safety factors and support cost estimates at feasibility stage. This workflow allows optimization of investigation campaigns, concentrating boreholes, scanline mapping, and laboratory testing in Najd-intense corridors to improve cost-benefit while reducing uncertainty in the most critical project areas.



This workflow transforms the Najd Fault System from background tectonic context into an active, quantitative design variable — the critical intellectual contribution of this research to engineering practice in the western Arabian Shield.

Rock-Mass Anisotropy Induced by Najd Shearing

The Najd Fault System makes the rock mass strongly anisotropic by imposing a persistent NW–SE shear-related fabric of joints, shear zones, and foliations that create direction-dependent strength, stiffness, and permeability. In engineering terms, the same rock type may behave as "good rock" when loaded or excavated in one direction and as "poor rock" when the same operations are conducted in a direction that mobilizes the Najd fabric. This directional dependency is not captured by conventional isotropic rock-mass models unless an anisotropic extension — such as ARMR or directional shear strength models — is explicitly adopted.

Structural / Geometrical Anisotropy Preferred NW–SE discontinuity orientation means block shapes, wedge geometries, and potential sliding planes are directionally organized. The rock mass behaves as a structurally layered medium at engineering scale even when intact rock is massive granite.	Mechanical Anisotropy Shear strength (friction angle, cohesion), deformation modulus, and Hoek–Brown parameters are all direction-dependent. Minima occur when the loading direction aligns with Najd-parallel joint or shear-zone planes. Isotropic models may overestimate stability in critical orientations.	Hydraulic Anisotropy Permeability is systematically higher parallel to Najd shear corridors and lower perpendicular to them. Groundwater flow, pore-pressure distribution, and weathering fronts are all concentrated along the structural fabric, compounding mechanical weakening on those planes.
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Discontinuity Properties Most Influenced by Najd Shearing

In order to translate the regional structural imprint of the Najd Fault System into quantitative rock-mass parameters, it is essential to identify which discontinuity properties are most strongly modified by Najd-related shearing. The following table summarizes the typical changes observed in key properties within Najd corridors and shows how these changes directly map into GSI and RMR ratings used in engineering design. These systematic modifications of discontinuity geometry and condition provide the critical bridge in the design chain from Regional Geology → Najd Structures → Joint Sets → Kinematics → GSI/RMR → Hoek–Brown → LEM → FS → Support Design.

Discontinuity Property	Typical Najd-Related Change	Direct Implication for GSI / RMR
Orientation & Persistence	Development of regionally persistent NW–SE shear zones and joint sets, steeply dipping and laterally continuous for hundreds of meters to kilometers	More unfavorable joint orientations in RMR and lower structural quality in GSI; continuous planes can daylight and control planar or wedge failures
Spacing / Frequency (RQD, Block Size)	Joint and fracture frequency increases within Najd corridors, reducing RQD and block size relative to adjacent unsheared rock	Lower RQD and smaller spacing classes in RMR; reduced blockiness component in GSI; direct degradation of rock-mass class
Surface Roughness & Waviness	Shear movement smooths and polishes joint surfaces, producing slickensides and reducing small-scale roughness and interlocking	Poorer joint-condition ratings in RMR and lower surface-condition component in GSI; reduced peak shear strength
Infill / Gouge & Alteration	Formation of clay-rich gouge and mineral infill; stronger alteration and wall-rock softening along major shear zones	Lower RMR joint-condition ratings and poorer GSI surface condition; weak fillings reduce friction angle and cohesion
Aperture & Connectivity	Damage zones increase effective aperture and fracture connectivity, especially along the shear fabric direction	Higher groundwater penalty in RMR; lower effective GSI where seepage is present; increased risk of hydraulic jacking and piping along shear corridors.